

Generalized Homology and the Eilenberg-Steenrod Axioms

Algebraic Topology I | Fall 2026 | September 30

Use Lecture 5. Give proofs and specify the source and target of each map. Work with spaces of CW homotopy type. Write $I = [0, 1]$. For every space W , write $\text{pr}_W : W \rightarrow *$ for the unique map to a point.

Exercise 1. Homotopy Invariant Functors

Let $\mathcal{C}$ be a category and $F : \text{Top} \rightarrow \mathcal{C}$ a functor on the spaces under consideration. Homotopy invariance means $F(f) = F(g)$ whenever $f, g : X \rightarrow Y$ are homotopic.

1. For $\epsilon \in \{0, 1\}$, let $i_\epsilon : X \rightarrow X \times I$ send x to (x, ϵ) , and let $p : X \times I \rightarrow X$ be projection. Construct a homotopy from $i_\epsilon p$ to the identity. Prove that a homotopy invariant F sends p to an isomorphism and that $F(i_0) = F(i_1) = F(p)^{-1}$.
2. Conversely, suppose $F(p)$ is an isomorphism for every X . Prove homotopy invariance by expressing the endpoints of $H : X \times I \rightarrow Y$ as Hi_0 and Hi_1 .
3. Let $f : X \rightarrow Y$ and $g : Y \rightarrow X$ satisfy $gf \simeq \text{id}_X$ and $fg \simeq \text{id}_Y$. Identify the inverse of $F(f)$. Explain why this argument applies to any target category.

Exercise 2. The Double Mapping Cylinder

For a span $X \xleftarrow{u} W \xrightarrow{v} Y$, set

$$M(u, v) = (X + (W \times I) + Y) / ((w, 0) \sim u(w), (w, 1) \sim v(w)).$$

Write $j_X : X \rightarrow M(u, v)$ and $j_Y : Y \rightarrow M(u, v)$ for the canonical maps.

1. Prove that a continuous map $M(u, v) \rightarrow Z$ is equivalent to maps $f : X \rightarrow Z$, $g : Y \rightarrow Z$ and a homotopy $H : fu \simeq gv$. Give the comparison map θ_H explicitly on all three pieces.
2. A square with these maps and chosen H is a homotopy pushout if θ_H is a homotopy equivalence. For the span $* \xleftarrow{\text{pr}_{S^0}} S^0 \xrightarrow{\text{pr}_{S^0}} *$, identify the ordinary pushout and the double mapping cylinder. Explain why the ordinary pushout square fails this condition.
3. Let $\alpha : W \rightarrow W'$, $\beta : X \rightarrow X'$ and $\gamma : Y \rightarrow Y'$ satisfy $\beta u = u' \alpha$ and $\gamma v = v' \alpha$. Define the induced map $\mu : M(u, v) \rightarrow M(u', v')$ on the three pieces and check the quotient identifications.

Exercise 3. Naturality and the Mayer-Vietoris Sequence

Let $A = (A_n)_{n \in \mathbb{Z}}$ satisfy homotopy invariance, additivity and the Mayer-Vietoris axiom, with the natural connecting maps of Lecture 5.

1. Explain why $j_{X,*} u_* = j_{Y,*} v_*$. Verify that the composite of the first two maps in

$$A_n(W) \xrightarrow{(u_*, -v_*)} A_n(X) \oplus A_n(Y) \xrightarrow{j_{X,*} + j_{Y,*}} A_n(M(u, v))$$

is zero. Explain what additional assertions are required for a long exact sequence.

2. For a homotopy pushout comparison $\theta_H : M(u, v) \rightarrow Z$, transport the entire Mayer-Vietoris sequence to $A_n(Z)$. Identify both the middle map and the connecting map δ_n^Z .
3. For the map of spans in Exercise 2, write the commutative square expressing naturality of $\delta_n : A_n(M(u, v)) \rightarrow A_{n-1}(W)$. Prove that it induces a morphism of the long exact sequences, including the signs in the first map.

Exercise 4. Coefficient Groups and Discrete Spaces

Write $G_n = A_n(*)$. The additivity axiom includes the empty family.

1. Prove $A_n(\emptyset) = 0$. For a discrete space X , describe the canonical isomorphism $\bigoplus_{x \in X} G_n \rightarrow A_n(X)$ and the homomorphism induced by a function $f : X \rightarrow Y$, including when different points have the same image.
2. For a nonempty contractible X , show that the map $\text{pr}_X : X \rightarrow *$ induces an isomorphism $A_n(X) \rightarrow G_n$. Show that all point inclusions $* \rightarrow X$ induce its inverse.
3. Let $B = (B_n)$ be another such theory and set $A_n(X) = B_{n-r}(X)$ for a fixed integer r . Define the connecting maps for A and verify the three axioms. If B satisfies the dimension axiom and $B_0(*) \neq 0$, determine when A satisfies it.

Exercise 5. Reduced Homology and Spheres

For a pointed nonempty space (X, x_0) , define $\tilde{A}_n(X) = \ker(\text{pr}_X)_*$. The point inclusion $s : * \rightarrow X$ sends the point to x_0 .

1. Prove the isomorphism $A_n(X) \rightarrow \tilde{A}_n(X) \oplus G_n$ given by $z \mapsto (z - s_*(\text{pr}_X)_*z, (\text{pr}_X)_*z)$. Write its inverse and prove naturality for maps preserving basepoints.
2. Choose one of the two points of S^0 as basepoint. Compute $\tilde{A}_n(S^0)$ using additivity. Identify $(\text{pr}_{S^0})_* : G_n \oplus G_n \rightarrow G_n$.
3. For $m \geq 1$, cover S^m by the complements of its north and south poles. Each member is contractible and the intersection is homotopy equivalent to S^{m-1} . Use the Mayer-Vietoris sequence to prove $\tilde{A}_n(S^m) \cong \tilde{A}_{n-1}(S^{m-1})$. Identify the map on the intersection's coefficient summand and explain how its kernel yields this isomorphism.
4. Deduce $A_n(S^m) \cong G_n \oplus G_{n-m}$ for all $m \geq 0$ and $n \in \mathbb{Z}$. State explicitly how the dimension axiom specializes this formula to ordinary integral homology, including $m = 0$.

Exercise 6. Relative Homology from a Mapping Cone

For an inclusion $i : X' \hookrightarrow X$, let C_i be the double mapping cylinder of $X \xleftarrow{i} X' \xrightarrow{\text{pr}_{X'}} *$, pointed at the cone vertex, and put $A_n(X, X') = \tilde{A}_n(C_i)$.

1. Apply Mayer-Vietoris to this span and split off the coefficient groups. Derive the long exact sequence

$$\cdots \rightarrow A_n(X') \xrightarrow{i_*} A_n(X) \rightarrow A_n(X, X') \rightarrow A_{n-1}(X') \rightarrow \cdots$$

Explain the sign in the map $A_n(X') \rightarrow A_n(X) \oplus G_n$ before splitting.

2. For $X' = \emptyset$, identify C_i and show $A_n(X, \emptyset) \cong A_n(X)$.
3. For $i = \text{id}_X$, construct a contraction of C_i to its cone vertex and deduce $A_n(X, X) = 0$.